

Software Engineering Project

"Student Expense Tracker"

A Software Engineering Project for Semester IV



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Academic Year: 2025-2026

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Introduction

Managing personal expenses is an important aspect of financial planning. Students often face difficulties tracking their daily expenditures, which can lead to overspending and poor budgeting. The Student Expense Tracker is a web-based application developed to help users record, categorize, analyze, and manage their expenses efficiently.

Project Overview

The Student Expense Tracker is a Flask-based web application that allows users to add, view, delete, and analyze their expenses. The system stores expense records in a SQLite database and provides reports, category-wise filtering, budget tracking, and data visualization features.

Main Features

- Add expenses
- Delete expenses
- View expense history
- Category-wise filtering
- Monthly expenditure reports
- Budget and savings estimation
- Data visualization using charts
- Export data to CSV/Excel

Objectives and Contribution

Objectives

- To help students manage daily expenses.
- To provide an easy-to-use budgeting tool.
- To generate monthly expenditure reports.
- To visualize spending patterns using charts.
- To improve financial awareness among students.

Contribution

This project demonstrates the practical implementation of:

- Web development using Flask
- Database management using SQLite
- CRUD operations
- Data visualization techniques
- User authentication concepts
- Financial data analysis

Problem Definition

Students frequently spend money on food, transportation, shopping, and entertainment without maintaining proper records. As a result, they often lose track of their spending habits and exceed their monthly budgets.

The problem addressed by this project is the lack of a simple and accessible tool for recording and monitoring daily expenses.

Topic Discussion

Expense Management

The application allows users to record expenses with the following details:

- Amount
- Category
- Description
- Date

Category Filtering

Expenses can be filtered based on categories such as:

- Food
- Transport
- Shopping
- Hostel
- Entertainment

Monthly Reports

The system calculates:

- Total monthly spending
- Highest spending category

- Estimated savings

Budget Tracking

Users can set a monthly budget, and the system automatically calculates remaining savings.

Data Visualization

Charts are generated using Chart.js to display:

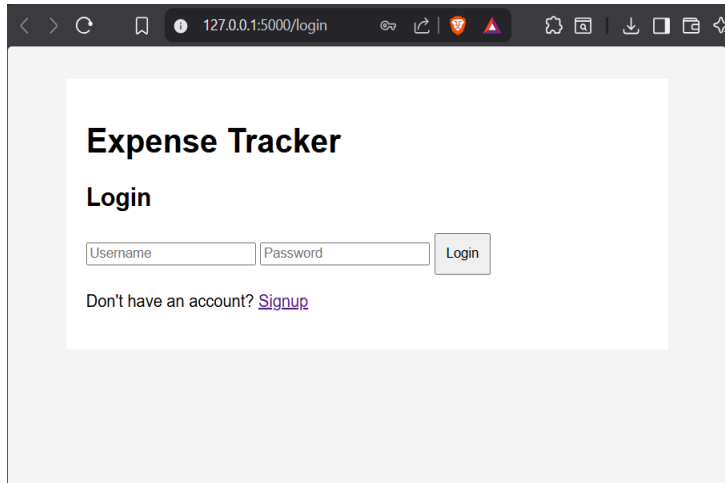
- Category-wise spending distribution
- Weekly spending trends

Data Export

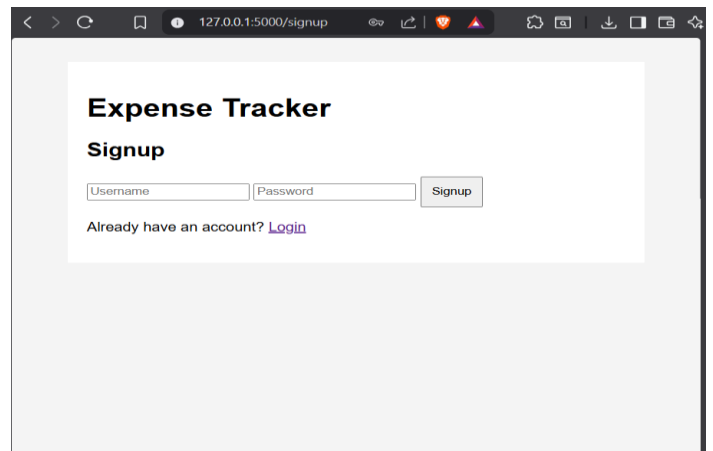
Users can export expense records in:

- CSV format
- Excel format

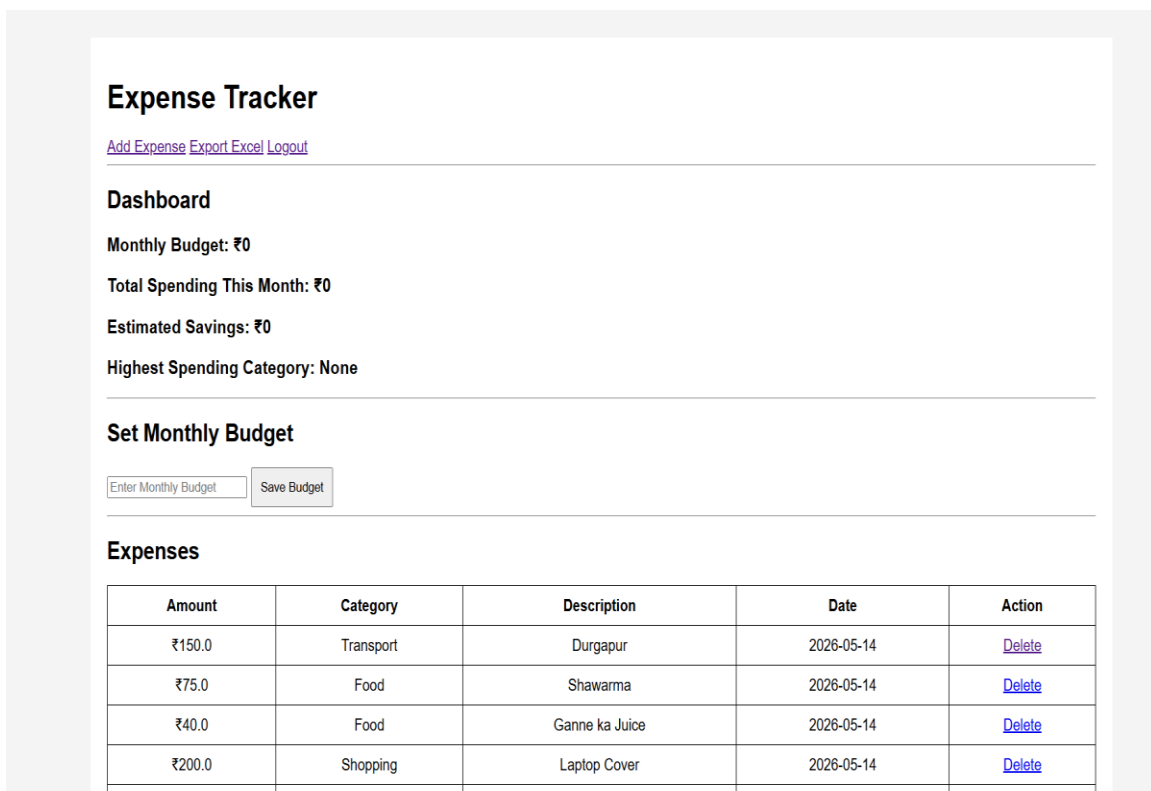
Some pic of the software



A screenshot of a web browser showing the login page of an 'Expense Tracker' application. The browser's address bar shows '127.0.0.1:5000/login'. The page has a light gray background. A white box in the center contains the title 'Expense Tracker' and the heading 'Login'. Below the heading are two input fields labeled 'Username' and 'Password', followed by a 'Login' button. At the bottom of the white box, there is a link that says 'Don't have an account? [Signup](#)'.



A screenshot of a web browser showing the signup page of an 'Expense Tracker' application. The browser's address bar shows '127.0.0.1:5000/signup'. The page has a light gray background. A white box in the center contains the title 'Expense Tracker' and the heading 'Signup'. Below the heading are two input fields labeled 'Username' and 'Password', followed by a 'Signup' button. At the bottom of the white box, there is a link that says 'Already have an account? [Login](#)'.



A screenshot of the 'Expense Tracker' dashboard. The page has a light gray background. At the top, there is a header section with the title 'Expense Tracker' and three links: 'Add Expense', 'Export Excel', and 'Logout'. Below the header is a section titled 'Dashboard' which displays four statistics: 'Monthly Budget: ₹0', 'Total Spending This Month: ₹0', 'Estimated Savings: ₹0', and 'Highest Spending Category: None'. Below the dashboard is a section titled 'Set Monthly Budget' with an input field labeled 'Enter Monthly Budget' and a 'Save Budget' button. At the bottom is a section titled 'Expenses' which contains a table with five columns: 'Amount', 'Category', 'Description', 'Date', and 'Action'. The table lists four expenses: ₹150.0 for Transport (Durgapur), ₹75.0 for Food (Shawarma), ₹40.0 for Food (Ganne ka Juice), and ₹200.0 for Shopping (Laptop Cover). Each row has a 'Delete' link in the 'Action' column.

Amount	Category	Description	Date	Action
₹150.0	Transport	Durgapur	2026-05-14	Delete
₹75.0	Food	Shawarma	2026-05-14	Delete
₹40.0	Food	Ganne ka Juice	2026-05-14	Delete
₹200.0	Shopping	Laptop Cover	2026-05-14	Delete

Experimental Setup and Tools Used

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection (optional)

Software Requirements

Operating System

- Windows
- Linux (Ubuntu/WSL)

Programming Language

- Python 3.x

Framework

- Flask

Database

- SQLite

Frontend Technologies

- HTML

- CSS
- JavaScript

Libraries

- Flask-SQLAlchemy
- Pandas
- Chart.js
- OpenPyXL

Development Environment

- Visual Studio Code
- Ubuntu WSL Terminal

Deployment Strategy

The project can be deployed using cloud hosting platforms such as:

- Render
- Railway
- PythonAnywhere

Deployment Steps

1. Upload project to GitHub.
2. Connect repository to hosting platform.
3. Configure Python environment.
4. Install dependencies from requirements.txt.
5. Deploy and test the application.

Results and Discussion

The developed application successfully performs all basic expense management operations.

Achieved Results

- Accurate storage of expense records.
- Efficient retrieval of financial data.
- Automatic calculation of monthly expenses.
- Budget tracking and savings estimation.
- Interactive visualization of spending patterns.
- Export functionality for external analysis.

Discussion

The application simplifies financial tracking for students and provides useful insights into spending behavior. The graphical representation helps users understand where most of their money is being spent.

Conclusion

The Student Expense Tracker is a simple yet effective financial management system designed for students. It provides essential features such as expense recording, budgeting, reporting, and visualization. The project demonstrates practical applications of web development, database management, and data analytics while addressing a real-world problem faced by many students.

Future enhancements may include user authentication, AI-based spending predictions, mobile responsiveness, and cloud database integration.

References

- [Flask Documentation](#)
- [SQLite Documentation](#)
- [Chart.js Documentation](#)
- [Pandas Documentation](#)
- [Python Official Documentation](#)
- [SQLAlchemy Documentation](#)